

The Impact of ChatGPT on Open-Source AI Model Adoption

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1 Selected AI Technology & Industry

Our group is currently examining Open-Source Large Language Models & AI Model Distribution through the Hugging Face AI model repository. We are focusing on AI research and development and examining whether the release of ChatGPT in 2022 has increased the adoption of LLM-based text generation models compared to non-LLM models.

This is significant from an economic perspective because AI is currently growing. Moreover, Hugging Face is a major platform that enables developers, researchers, and firms to access and distribute AI models. If ChatGPT increased demand for large language models, this can show that the AI breakthrough has shifted investment and innovation toward generative AI. It can also help explain how firms allocate their resources to AI tools that can improve productivity, automation, and product development.

Open-source AI model distribution may lead to more advanced AI tools in the future that can be accessible to startups, researchers, and smaller organizations that do not have the resources to build models from scratch. This can foster innovation, but there are concerns about job displacement, AI accountability, and misinformation. The rise of LLMs across different organizations can change workflows by increasing the use of AI for tasks such as writing, coding, and data analysis.

We are looking to answer the following questions:

- Did the release of ChatGPT in November 2022 increase the adoption of LLM-based text generation models on Hugging Face?
- Did LLM-based models experience higher growth in downloads and engagement compared to non-LLM models after ChatGPT's release?
- What do changes in Hugging Face model activity suggest about broader economic investment and attention in generative AI?

2 Preliminary Data and Summary Statistics

For this project, we use data from the Hugging Face model repository [1], which is a major platform for sharing open-source AI models. The dataset is accessed through the Hugging Face API, which provides information about different models. Key variables include model name, number of downloads, number of likes, and model category (pipeline tag). We focus on models in the *text-generation* category as a proxy for LLM-based models and compare them with non-LLM models, such as image classification models.

We collect the top 100 models in each category, sorted by number of downloads, to better reflect real-world adoption. We then calculate summary statistics, including average (mean) downloads and median downloads.

Table 1: Summary Statistics by Model Category

Model Category	Number of Models	Average Downloads	Median Downloads
LLM-Based Models: Text Generation	100	2,793,651	141,375
Non-LLM Models: Image Classification	100	894,351	134,758

These results show that LLM-based models have a much higher average level of adoption compared to non-LLM models. The average number of downloads for text generation models is 2.79 million, compared to 894,351 for image classification. However, the median values are relatively close—for text generation, the median download is 141,375 and 134,758 for image classification. This suggests that the difference in averages may be driven by a small number of highly popular models, indicating a skewed distribution where a few models account for a large share of total downloads.

This pattern is important for our project, as it may suggest that the rise of generative AI has increased attention and adoption for certain high-profile LLM models, but the effect may not be evenly distributed across all models in the category.

For this project, we focus mainly on adoption rates, using Hugging Face downloads and likes as indicators of model usage and community interest [2]. These metrics have helped us assess

whether LLM-based text generation models received more attention than non-LLM models following the release of ChatGPT. Regarding market size, employment, or productivity, our data does not necessarily cover those. However, higher adoption can suggest growing investment in generative AI and potential organizational changes—for example, AI usage increasing the need for coding, customer support, and workflow automation [3].

3 Limitations

There are several limitations in our data. First, the Hugging Face API does not provide a perfect classification of LLM models, so we use the *text-generation* category as a proxy, which may not include all relevant models. Second, the number of downloads may not fully represent actual usage, since downloading a model does not guarantee that it is actively used. Third, the data is highly skewed, with a few models showing extremely large download counts, which can affect the average. Finally, external factors such as general trends in AI development and media attention may also influence adoption, making it difficult to isolate the specific impact of ChatGPT [4].

4 GitHub Copilot Reflection

For this project, we used Copilot to help draft the Python script that queries the Hugging Face API. The assistant generated the request structure, including the endpoint URL and the parameter dictionary with `pipeline_tag`, sort order, and result limit. It also helped set up the logic for converting the JSON response into a pandas DataFrame, removing null values, and calculating the average and median downloads shown in our preliminary results table.

We also used the assistant when setting up the treatment-versus-control comparison. It suggested filtering on Hugging Face’s `pipeline_tag` field, using *text-generation* for the LLM-

based treatment group and *image-classification* for the non-LLM control group. That approach let us pull two parallel datasets using the same code structure, which made the comparison easy to run and simple to expand if we decide to include more task categories later.

The assistant greatly accelerated our research process, especially during the setup phase. Instead of spending hours digging through the Hugging Face API documentation, we ended up with a working scraper and clean output in just a few minutes. That let us spend more time interpreting the results instead of getting stuck writing repetitive setup code.

At the same time, it added some complications that were not obvious at first. The initial script ran without errors and produced clean numbers, but it did not include the `createdAt` field we needed for our Difference-in-Differences design. That experience made it clear that AI-generated code should be treated as a draft that still needs to be reviewed carefully, not something ready to use right away.

By the end, we saw AI less as an expert and more as a fast, helpful collaborator that still needs humans to set the goals and double-check the results. The same pattern we observed in our own workflow—where AI speeds up routine tasks but still needs humans for the harder judgment calls—is probably happening across the open-source AI ecosystem we are studying [5]. Instead of doing repetitive execution, humans end up focusing more on defining the problems, checking the outputs, and making the decisions that AI still cannot make on its own.

5 References

- [1] Hugging Face. (2023). *Hugging Face model hub*. <https://huggingface.co/models>
- [2] Wolf, T., Debut, L., Sanh, V., et al. (2020). Transformers: State-of-the-art natural language processing. *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing: System Demonstrations*, 38–45. <https://aclanthology.org/2020.emnlp-demos.6>
- [3] Brynjolfsson, E., Li, D., & Raymond, L. (2023). *Generative AI at work*. NBER Working Paper No. 31161. <https://www.nber.org/papers/w31161>
- [4] Bommasani, R., Hudson, D. A., Aditi, E., et al. (2021). *On the opportunities and risks of foundation models*. Stanford CRFM. <https://crfm.stanford.edu/assets/report.pdf>
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